

Certification Project

Microsoft .NET Framework

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Introduction, aim, purpose

Introduction

This certification project is a small Library Management System built with **ASP.NET Web Forms (C#)** and **SQL Server**.

The idea is based on the scenario where a college principal wants to move the library records from paper to a basic web application. As a learner, my goal was not to build a perfect commercial system, but to practice and apply the topics covered in the training:

- Web Forms and master pages
- ADO.NET and SQL Server
- State management, security and web services concepts

Aim and purpose

The main aim of this project is to create a working prototype that allows a librarian to manage books and students, and to issue and return books with penalties calculated for late returns.

The purpose of the system is:

- to give the **admin (librarian)** a simple interface to maintain branches, publications, books and students, and to track issues, returns and penalties;
- to give the **student** a way to log in, view and update their details, and see their own borrowing history and penalties;
- to demonstrate my understanding of the .NET Framework, ASP.NET Web Forms, database design, and basic CRUD operations.

Additional Notes

- All **SQL scripts** (database creation, table creation, sample data) **and the full project source code** are compressed and attached with the submission.
- The project's emphasis is **functionality over aesthetics**.
A simple default CSS style was used; a more advanced, responsive UI (Bootstrap, custom design, icons, etc.) could be added in the future.
- The **Book** table stores basic information (title, author, branch, publication, price, quantity). Additional fields such as **ISBN, edition, year, category, language** could be added to improve real-world book cataloging.
- Error handling, form validation, and security are implemented at a basic level suitable for a prototype, but can be further enhanced.

Database and Table creation

The application uses a single SQL Server database named **LibraryDB**.

The database was designed first, and an ER diagram was created to show the tables and their relationships (diagram attached in the report).

Main tables

- **Branch** – stores the list of branches/departments (e.g. Computer Science, Mechanical Engineering).
- **Publication** – stores publishers of books (e.g. Pearson, McGraw-Hill).
- **AdminUser** – stores librarian login details. There is no relation to other tables; it is only used for authentication.
- **Student** – stores student details such as name, branch, contact information and login credentials. It has a foreign key to **Branch**.
- **Book** – stores book information (title, author, publication, branch, price, quantity). It has foreign keys to **Branch** and **Publication**.
- **IssueBook** – stores the history of issued books. It links **Student** and **Book** through foreign keys and keeps **IssueDate**, **DueDate**, **ReturnDate** and **Penalty**. This table is the core for reports and penalty calculations.

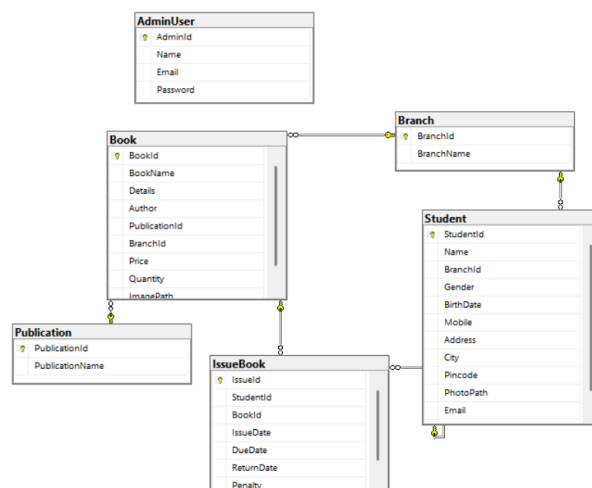
Relationships (from the DB diagram)

- One **Branch** can have many **Students** and many **Books**.
- One **Publication** can have many **Books**.
- One **Student** can have many **IssueBook** records.
- One **Book** can appear in many **IssueBook** records over time.

These relationships enforce data integrity (for example, a book cannot be issued to a non-existent student).

After creating the tables, some initial data was inserted:

- One admin user (Librarian) for logging in as admin.
- Sample branches, publications and books for different departments.
- A few test students to simulate real usage and to test login, issue, return and penalty features.



T-SQL script

The T-SQL script used to create the database, tables and seed data is shown below:

```
CREATE DATABASE LibraryDB;

USE LibraryDB;

-- 1. Branch table
CREATE TABLE Branch (
    BranchId INT IDENTITY PRIMARY KEY,
    BranchName NVARCHAR(100) NOT NULL
);

-- 2. Publication table
CREATE TABLE Publication (
    PublicationId INT IDENTITY PRIMARY KEY,
    PublicationName NVARCHAR(200) NOT NULL
);

-- 3. Student table
CREATE TABLE Student (
    StudentId INT IDENTITY PRIMARY KEY,
    Name NVARCHAR(100),
    BranchId INT FOREIGN KEY REFERENCES Branch(BranchId),
    Gender NVARCHAR(10),
    BirthDate DATE,
    Mobile NVARCHAR(20),
    Address NVARCHAR(200),
    City NVARCHAR(100),
    Pincode NVARCHAR(10),
    PhotoPath NVARCHAR(200),
    Email NVARCHAR(100) UNIQUE,
    Password NVARCHAR(50)
);

-- 4. Admin table (Librarian accounts)
CREATE TABLE AdminUser (
    AdminId INT IDENTITY PRIMARY KEY,
    Name NVARCHAR(100),
    Email NVARCHAR(100) UNIQUE,
    Password NVARCHAR(50)
);

-- 5. Books
CREATE TABLE Book (
    BookId INT IDENTITY PRIMARY KEY,
    BookName NVARCHAR(200),
    Details NVARCHAR(500),
    Author NVARCHAR(200),
    PublicationId INT FOREIGN KEY REFERENCES Publication(PublicationId),
    BranchId INT FOREIGN KEY REFERENCES Branch(BranchId),
    Price DECIMAL(10,2),
    Quantity INT,
    ImagePath NVARCHAR(200)
);

-- 6. IssueBook table (issue + return)
CREATE TABLE IssueBook (
    IssueId INT IDENTITY PRIMARY KEY,
    StudentId INT FOREIGN KEY REFERENCES Student(StudentId),
    BookId INT FOREIGN KEY REFERENCES Book(BookId),
    IssueDate DATE,
    DueDate DATE,
```

```
ReturnDate DATE NULL,  
Penalty DECIMAL(10,2) DEFAULT 0  
);
```

```
INSERT INTO AdminUser (Name, Email, Password)  
VALUES ('Librarian', 'admin@library.com', 'admin123');
```

```
USE LibraryDB;
```

```
-- Create one branch:
```

```
INSERT INTO Branch (BranchName) VALUES ('Computer Science');
```

```
-- Computer Science (BranchId = 1)
```

```
INSERT INTO Book (BookName, Details, Author, PublicationId, BranchId, Price, Quantity, ImagePath)  
VALUES  
( 'Introduction to Algorithms', 'Algorithm design and analysis', 'Thomas H. Cormen', 1, 1, 89.99, 10, NULL),  
( 'Clean Code', 'Writing maintainable code', 'Robert C. Martin', 3, 1, 45.50, 7, NULL),  
( 'Operating System Concepts', 'OS principles and design', 'Abraham Silberschatz', 2, 1, 79.00, 5, NULL);
```

```
-- Electrical Engineering (BranchId = 3)
```

```
INSERT INTO Book (BookName, Details, Author, PublicationId, BranchId, Price, Quantity, ImagePath)  
VALUES  
( 'Electrical Machines', 'Concepts of AC/DC machines', 'P.S. Bimbhra', 4, 3, 55.00, 6, NULL),  
( 'Power Electronics', 'Power devices and systems', 'Muhammad H. Rashid', 2, 3, 65.00, 8, NULL);
```

```
-- Mechanical Engineering (BranchId = 2)
```

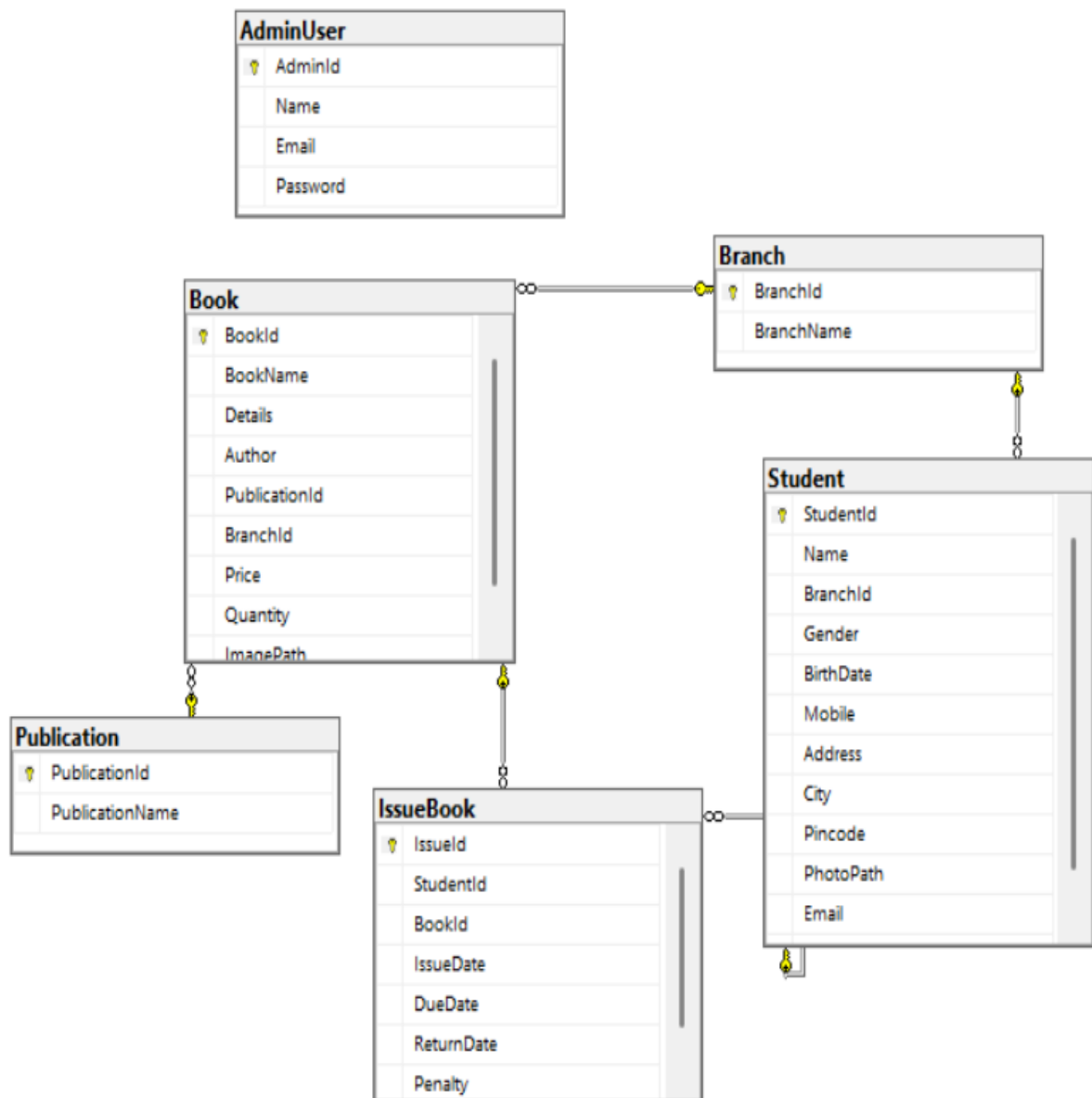
```
INSERT INTO Book (BookName, Details, Author, PublicationId, BranchId, Price, Quantity, ImagePath)  
VALUES  
( 'Strength of Materials', 'Material strength concepts', 'R.K. Bansal', 1, 2, 40.00, 12, NULL),  
( 'Thermodynamics', 'Engineering thermodynamics', 'Yunus A. Cengel', 5, 2, 75.00, 4, NULL);
```

```
-- Create a sample student
```

```
INSERT INTO Student  
(Name, BranchId, Gender, BirthDate, Mobile, Address, City, Pincode,  
PhotoPath, Email, Password)  
VALUES  
( 'Test Student', 1, 'M', '2000-01-01', '1234567890',  
'Test Address', 'Test City', '12345',  
NULL, 'student@library.com', 'stud123');
```

```
-- Add more Students
```

```
INSERT INTO Student (Name, BranchId, Gender, BirthDate, Mobile, Address, City, Pincode, PhotoPath, Email, Password)  
VALUES  
( 'Alice Johnson', 1, 'F', '2001-02-14', '9876543210', '123 Main St', 'New York', '10001', NULL, 'alice@library.com', 'alice123'),  
( 'Ravi Kumar', 2, 'M', '2000-05-22', '9123456780', '22 Hill Road', 'Mumbai', '400050', NULL, 'ravi@library.com', 'ravi123'),  
( 'Sarah Lee', 3, 'F', '2002-03-10', '9988776655', '45 Elm Street', 'Toronto', 'M4B1B3', NULL, 'sarah@library.com', 'sarah123');
```



Logging

Admin

Library Login

https://local...

Library Management – Login

Email:

admin@library.com

Password:

Login

Admin Home Dashboard

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Admin Home

Welcome, Librarian

Library Summary

Item	Count
Total Branches	8
Total Publications	6
Total Books	7
Total Students	4
Books Issued (Active)	8
Overdue Books	2

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Add Branch

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Add Branch

Branch name:

Add

ID	Branch	
1	Computer Science	Edit Delete
2	Electrical Engineering	Edit Delete
3	Mechanical Engineering	Edit Delete
4	Civil Engineering	Edit Delete
5	Electronics & Communication	Edit Delete
6	Information Technology	Edit Delete
7	Business Administration	Edit Delete
8	Mathematics	Edit Delete

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Add Publication

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Add Publication

Publication name:

Add

ID	Publication	
1	Pearson	Edit Delete
2	McGraw-Hill	Edit Delete
3	O'Reilly Media	Edit Delete
4	Springer	Edit Delete
5	Wiley	Edit Delete
6	Oxford University Press	Edit Delete

Add Books

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Add Book

Book Name

Details

Author

Publication

Pearson

Branch

Computer Science

Price

Quantity

Add Book

BookId	BookName	Author	PublicationName	BranchName	Price	Quantity
1	Introduction to Algorithms	Thomas H. Cormen	Pearson	Computer Science	89.99	9
2	Clean Code	Robert C. Martin	O'Reilly Media	Computer Science	45.50	7
3	Operating System Concepts	Abraham Silberschatz	McGraw-Hill	Computer Science	79.00	4
4	Electrical Machines	P.S. Bimbhra	Springer	Mechanical Engineering	55.00	5
5	Power Electronics	Muhammad H. Rashid	McGraw-Hill	Mechanical Engineering	65.00	7
6	Strength of Materials	R.K. Bansal	Pearson	Electrical Engineering	40.00	10
7	Thermodynamics	Yunus A. Cengel	Wiley	Electrical Engineering	75.00	4

Issue Book

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Issue Book

Student

Sarah Lee

▼

Publication

McGraw-Hill

▼

Book

Operating System Concepts

▼

Due date (days from today)

14

Issue Book

Book issued successfully.

Currently Issued Books

Filter by student:

All Students

 ▼

IssueId	Student	BookName	IssueDate	DueDate	ReturnDate
6	Sarah Lee	Electrical Machines	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
7	Ravi Kumar	Thermodynamics	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
8	Ravi Kumar	Strength of Materials	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
9	Sarah Lee	Power Electronics	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
10	Sarah Lee	Introduction to Algorithms	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
11	Sarah Lee	Operating System Concepts	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
3	Alice Johnson	Operating System Concepts	12/7/2025 12:00:00 AM	12/21/2025 12:00:00 AM	
4	Alice Johnson	Introduction to Algorithms	11/21/2025 12:00:00 AM	12/6/2025 12:00:00 AM	
5	Sarah Lee	Strength of Materials	11/20/2025 12:00:00 AM	12/5/2025 12:00:00 AM	

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Return Book

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Return Book

Student

Alice Johnson

 ▼

Issue ID	Book	Issue Date	Due Date	
4	Introduction to Algorithms	2025-11-21	2025-12-06	Return
3	Operating System Concepts	2025-12-07	2025-12-21	Return

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Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Return Book

Student

Alice Johnson

 ▼

Book returned. Penalty: \$5.00.

Issue ID	Book	Issue Date	Due Date	
3	Operating System Concepts	2025-12-07	2025-12-21	Return

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Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Return Book

Student

Ravi Kumar

Book returned. No penalty.

Issue ID	Book	Issue Date	Due Date	
8	Strength of Materials	2025-12-07	2025-12-21	Return

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Penalty Report

Library Management

Library Management System

Admin: Librarian | [Logout](#)

[Admin Home](#) | [Add Branch](#) | [Add Publication](#) | [Add Book](#) | [Issue Book](#) | [Return Book](#) | [Penalty Report](#)

Penalty Report

Branch

All Branches

Student

All Students

Student	Book	Issue Date	Due Date	Return Date	Penalty
Alice Johnson	Introduction to Algorithms	2025-11-21	2025-12-06	2025-12-07	\$5.00
Sarah Lee	Strength of Materials	2025-11-20	2025-12-05	2025-12-07	\$10.00

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Student

Library Login

https://localhost:44302/Login.aspx

Library Management – Login

Email:

alice@library.com

Password:

Login

Student Home Dashboard

Library Management System

Student: Alice Johnson | [Logout](#)

[Student Home](#) | [My Account](#) | [My Report](#)

Student Home

Welcome, Alice Johnson

My Library Summary

Item	Value
Books currently issued	1
Total books borrowed	2
Total penalty (all time)	\$5.00

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Student Profile Details

Library Management

https://localhost:44302/MyAccou... Summarize

Library Management System

Student: Alice Johnson | [Logout](#)

[Student Home](#) | [My Account](#) | [My Report](#)

My Account

Profile Details

Name

Alice Johnson

Branch

Computer Science

Mobile

9876543210

Address

123 Main st

City

New York

Pincode

10001

Email

alice@library.com

Save Changes

Change Password

Old Password

New Password

Confirm New Password

Change Password

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Student Report

Conclusion

The system allows the admin to manage books, branches, publications, students, and circulation (issue/return), while students can log in, view reports, and update their profile. The database design enforces relationships and integrity through foreign keys, and all major operations are handled through ADO.NET.

Although the UI design is simple and the book information can be expanded, the system fulfills its educational purpose: demonstrating the application of .NET Framework skills learned throughout the course. The attached SQL scripts and full project code allow the project to be installed, tested, and extended as needed.